

ENGINEERING REBOOT 2026 - WEEK 08

OBJECT-ORIENTED DESIGN: INHERITANCE - POLYMORPHISM - EXTENSIBLE SYSTEMS

DAY 36 INHERITANCE FUNDAMENTALS FOCUS - Base & derived classes - Inherited state & behavior - is-a relationships BUILD Employee Hierarchy OUTCOME Create meaningful subclasses.	DAY 37 OVERRIDING & super() FOCUS - Method overriding - Constructor specialization - Shared vs. specialized behavior BUILD Specialized Employee Behavior OUTCOME Extend parent behavior cleanly.	DAY 38 POLYMORPHISM FOCUS - Common behavior - Dynamic dispatch - Mixed object collections BUILD Polymorphic Payroll OUTCOME Process types through one operation.	DAY 39 COMPOSITION vs. INHERITANCE FOCUS - is-a vs. has-a - Responsibility placement - Design justification BUILD Employee Management Model OUTCOME Choose the right relationship.	DAY 40 INTEGRATION PROJECT FOCUS - Inheritance + overriding - Polymorphism + composition - Encapsulation BUILD Employee Payroll Model OUTCOME Integrate the week's design skills.
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WEEK 08 ENGINEERING RULE

Use inheritance for meaningful specialization. Use composition for meaningful collaboration. Let objects own their behavior.

SCOPE DISCIPLINE: approximately 1 hour per session - learn the concept - demonstrate it - capture evidence - move forward.